

Linear Function Word Problems

Directions: Write a linear function for each situation. Show all work. No calculators.

1. A taxi charges a \$4 starting fee plus \$2.50 per mile. Write a linear function for the total cost after x miles. Then find the total cost after 8 miles.
2. A bicycle rental shop charges a \$12 rental fee plus \$6 per hour. Write a linear function for the total cost after x hours. Then find the total cost after 5 hours.
3. A cleaning company charges a \$75 service fee plus \$18 per hour. Write a linear function for the total cost after x hours of work. Then find the total cost after 4 hours of work.

4. A phone battery is at 96% and decreases by 7 percentage points each hour. Write a linear function for the battery percentage after x hours. Then find the battery percentage after 6 hours.
5. A water tank contains 800 gallons and drains at 35 gallons per minute. Write a linear function for the amount of water remaining after x minutes. Then find the amount of water remaining after 12 minutes.
6. A candle is 24 centimeters tall and burns down 1.5 centimeters per hour. Write a linear function for the candle height after x hours. Then find the candle height after 8 hours.
7. A student already has \$45 saved and adds \$12 each week. Write a linear function for the total amount saved after x weeks. Then find the total amount saved after 9 weeks.

8. A parking garage charges a \$5 entry fee plus \$3 per hour. Write a linear function for the total parking cost after x hours. Then find the total parking cost after 7 hours.
9. The temperature is 68°F at noon and decreases by 4°F each hour. Write a linear function for the temperature x hours after noon. Then find the temperature 5 hours after noon.
10. A fundraiser begins with \$150 in donations and collects \$25 for each ticket sold. Write a linear function for the total amount raised after x tickets are sold. Then find the total amount raised after 14 tickets are sold.